

Ariful Islam

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OVERVIEW

Recent B.Sc. (Engineering) graduate in Information and Communication Engineering at Pabna University of Science and Technology; four-year degree completed; final result published. CGPA 3.76/4.00 (Graduated). Research experience spans computer vision, deep learning, smart agriculture, biomedical imaging, and deepfake forensics. First author of a peer-reviewed IEEE paper and a published Mendeley Data dataset. Selected as one of only seven students from PUST for the Sakura Science Exchange Program in Japan, with additional experience in ICT instruction.

EDUCATION

B.Sc. (2026) **Bachelor of Science (Engineering) in Information and Communication Engineering**, Department of ICE, Pabna University of Science and Technology (PUST), Bangladesh.

CGPA: 3.76/4.00 (Graduated); four-year degree completed in 2026; final 8th-semester examination completed; result published.

Undergraduate Thesis: “Development and Evaluation of a YOLO-Based Mango Fruit Health Detection Framework Using the MangoFruitBD Dataset.”

Thesis-based manuscript submitted to *Computers and Electronics in Agriculture*.

HSC (2019) **Higher Secondary Certificate (Science)**, Ahmedpur College (1), Natore. GPA: 5.00/5.00.

SSC (2017) **Secondary School Certificate (Science)**, Hoybotpur Multi High School, Natore. GPA: 5.00/5.00.

CERTIFICATIONS & INTERNATIONAL EXPOSURE

- **Sakura Science Exchange Program, Japan** (25 Oct – 2 Nov 2025) – **Selected as one of only seven students from Pabna University of Science and Technology (PUST)** for the Japan exchange program. Completed an intensive international academic exchange with structured Deep Learning training from foundational concepts through U-Net architectures, together with cross-cultural academic engagement.
- **Unlocking Research: From Concept to Publication, Solver Green** (2025) – Completed a 14-session research-development programme covering research-question formulation, methodology, academic writing, publication ethics, and the publication process.
- **SQL (Basic) Certification** – HackerRank.

EXPERIENCE

ICT Instructor, Redox Educare, Pabna

2023 – Jun 2026

- Delivered structured ICT instruction in digital systems, networking, HTML/CSS, introductory C programming, and database concepts; prepared notes, practice materials, revision sessions, and

assessments.

ICT Instructor, Success Point Pabna

2022 – 2024

- Taught core ICT concepts through worked examples and guided practical exercises in digital logic, networking, web fundamentals, C programming, and database fundamentals.

Peer Reviewer, AETiC Journal

Academic Service

SELECTED RESEARCH EXPERIENCE

MangoFruitBD & YOLO-Based Mango Health Detection

2025 – 2026

- Co-developed a field-collected dataset of 1,310 RGB orchard images across Healthy, Anthracnose, Alternaria, and Scab classes.
- Contributed to image acquisition, bounding-box annotation review, dataset organization and splitting, and development/evaluation of a YOLO-based framework under real-field variability.

Deepfake Detection & Forensics

2026

- Investigated graph-attentional relational reasoning for deepfake forensics and comparative CNN, Transformer, and recurrent-model performance under degraded-image conditions.

PUBLICATIONS

Submitted Journal Manuscript:

1. **A. Islam** et al., “Development and Evaluation of a YOLO-Based Mango Fruit Health Detection Framework Using the MangoFruitBD Dataset.” Submitted to *Computers and Electronics in Agriculture*.

Peer-Reviewed Conference Papers:

1. **A. Islam**, S. K. Ray, S. A. Oishi et al., “A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images,” *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, 2026. [10.1109/QPAIN69676.2026.11545784](https://doi.org/10.1109/QPAIN69676.2026.11545784) [First Author].
2. S. K. Ray, **A. Islam**, M. C. Chanda et al., “Deep Learning Based Lung Image Segmentation Using XR-U-Net,” *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 2024. [10.1109/ICCIT64611.2024.11022346](https://doi.org/10.1109/ICCIT64611.2024.11022346).
3. M. A. Miya, M. H. Mohona, M. A. Hossain, M. B. A. Z. Shammo, **A. Islam** et al., “Remote Health Monitoring via PPG Signal Processing and Machine Learning Using Mobile Devices,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, 2025. [10.1109/ICCIT68739.2025.11491546](https://doi.org/10.1109/ICCIT68739.2025.11491546).
4. S. H. Turza, A. Mahdi, M. I. S. Shad, M. H. Kabir, and **A. Islam**, “Fake News Detection in Bangla: A Transformer Ensemble Framework for Robust Classification,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, 2025. [10.1109/ICCIT68739.2025.11491228](https://doi.org/10.1109/ICCIT68739.2025.11491228).

Published Dataset:

1. **A. Islam**, S. A. Oishi, M. M. Hasan et al., “MangoFruitBD: A Bounding-Box Annotated Image Dataset for Detecting Healthy and Diseased Mango Fruits in Bangladeshi Orchards,” *Mendeley Data*, Version 1, 2026. [10.17632/bhrz29mkmr.1](https://doi.org/10.17632/bhrz29mkmr.1) [First Author].

Manuscript Under Review:

1. “Precision Weeding in Agriculture: A Review of Deep Learning Methods for Weed Detection.”

RESEARCH, PROGRAMMING & COMPUTER SKILLS

Research & Data	Field image acquisition, bounding-box annotation, dataset curation and splitting, model training and evaluation, literature review, scholarly writing.
AI & Computer Vision	PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas, Matplotlib.
Programming	Python, C, C++, Java, JavaScript, PHP, SQL, MATLAB.
Systems & Tools	Jupyter Notebook, Kaggle, Visual Studio Code, SQL Server Management Studio, Proteus, mikroC, Cisco Packet Tracer.
Academic Coursework	Artificial Intelligence and Robotics, Neural Networks, Natural Language Processing, Digital Image and Speech Processing, Digital Signal Processing, DBMS, Web Programming, Software Analysis and System Testing, Wireless Communications, Computer Architecture and Microcontroller Design.

PROJECTS

Database	Hospital Management System – SQL/DBMS academic project.
Embedded	Automatic Fan Speed Control System – Microcontroller and hardware interfacing project.

ACHIEVEMENTS

- **Dutch-Bangla Bank Limited (DBBL) Scholarship** – Merit-based scholarship awarded on the basis of Higher Secondary Certificate board results.
- **Mini Bar Football Tournament Champion (Team Captain)** – Intra-Department Sports Week “ICE Fiesta 2026.”
- **Two-time Card-29 Champion** – Intra-Department Sports Week “ICE Fiesta 2025” and “ICE Fiesta 2023.”
- **Debate Champion** – Intra-Department Sports Week “ICE Fiesta 2025.”
- **Two-time Programming Contest Runner-up** – Intra-Department Sports Week “ICE Fiesta 2025” and “ICE Fiesta 2024.”
- **Carrom Champion** – Intra-Department Sports Week “ICE Fiesta 2024.”
- **Top 3 Campus Coordinator, Q1 2025** – Recognized by YUNet - International Youth Development Organization.

TALKS & PRESENTATIONS

Jun 2026	Presented “GR-ACE Net: A Hybrid Graph-Attentional Framework with Global Relational Reasoning for Deepfake Forensics” as first author at PECCII 2026, 17–18 June 2026, PUST, Pabna, Bangladesh.
Jun 2026	Presented “A Comparative Study of CNN, Transformer, and Recurrent Models for Deepfake Detection on Degraded Images” as second author at PECCII 2026, 17–18 June 2026, PUST, Pabna, Bangladesh.

VOLUNTEERING & LEADERSHIP

- President – Natore Students Welfare Association, PUST (2025 – 2026).
- Vice President – PUST Career and Entrepreneurship Club (2025 – 2026).
- Joint Organizing Secretary – Solver Green, PUST (2025 – 2026).
- Campus Coordinator – YUNet - International Youth Development Organization (2025); certified as one of the Top 3 Campus Coordinators for Q1 2025.

RESEARCH INTERESTS

Image-based AI; computer vision and deep learning; smart agriculture and precision farming; biomedical image segmentation and healthcare AI; low-resource NLP.

LANGUAGES

- **Bangla** – Native language.
- **English** – Medium of instruction throughout the B.Sc. (Engineering) program; academic and professional working proficiency.

REFERENCES

Dr. Md. Anwar Hossain

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